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METHOD FOR AUTOMATICALLY CONTROLLING AN AIRCRAFT IN THE EVENT OF A FIRE IN THE ENGINE ZONE AND AN AIRCRAFT

Abstract

A method for controlling an aircraft comprising at least two engines, each engine being arranged in its own engine compartment, each engine being connected to a fuel supply circuit provided with its own fuel supply cut-off for cutting off a fuel supply, the aircraft comprising at least one fire detector in each engine compartment, the aircraft comprising at least one first extinguisher and one second extinguisher. An automatic assistance phase comprises, in the event of fire being detected in an engine compartment, automatically and with an avionics system, activating the fuel supply cut-off of the engine present in the compartment concerned, then triggering the first extinguisher, then triggering the second extinguisher in predetermined respective conditions.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority to French Patent Application No. FR 24 02011 filed on Feb. 29, 2024, the disclosure of which is incorporated in its entirety by reference herein.

TECHNICAL FIELD

[0002] The present disclosure relates to a method for automatically controlling an aircraft in the event of a fire in the engine zone and an aircraft applying this method.

BACKGROUND

[0003] An aircraft may comprise one or more engines.

[0004] For example, a conventional helicopter may comprise several engines that together set a main rotor in motion via a power transmission system. Each engine is also positioned in an engine compartment. Fire detectors are arranged in each engine compartment to detect the possible presence of a fire. Such an aircraft may also comprise two extinguishers. Each extinguisher may transfer an anti-fire agent, such as a halon, to each engine compartment. The aircraft also comprises one fuel shut-off valve per engine for cutting off the fuel supply to the associated engine.

[0005] In the event of a fire, the pilot must interpret the alarm generated by a fire detector, identify the engine concerned by the alarm, close the fuel shut-off valve in order not to supply fuel to the fire, trigger an extinguisher and check that the alarm stops. This procedure is complicated because the pilot needs to interpret the situation depending on the alarm that is generated. This procedure entails a heavy workload at a particularly anxious time. The stress the pilot is under may lead to an incident due to a misinterpretation of the situation. In extreme cases, a pilot may even ignore the alarm triggered by a fire detector or seek to perform an emergency landing, forgetting to extinguish the fire.

[0006] On some aircraft, engine shutdown may be assisted. On some aircraft, a single button may be used to trigger the two extinguishers.

[0007] Documents CN 109 533 348 A, US 2019/126082 A1, FR 3 130 751 A1, and “chapter 17: fire protection systems”, Aug. 28, 2016 (2016-08-28), XP002791327, retrieved from the Internet: URL: http://www.sweethaven02.com/aviation/mainthandbook/ama_ch17.pdf are also known.

SUMMARY

[0008] An object of the present disclosure is thus to propose a method for reducing the workload of a pilot in order to reduce the risks of human error in the event of a fire being detected in an engine compartment of an aircraft.

[0009] The disclosure therefore relates to a method for controlling an aircraft comprising at least two engines, each engine being arranged in its own engine compartment, each engine being connected to a fuel supply circuit provided with its own fuel supply cut-off for cutting off a fuel supply, said aircraft comprising at least one fire detector in each engine compartment, said aircraft comprising at least one first extinguisher and one second extinguisher.

[0010] The control method comprises: [0011] detecting, with at least one of the fire detectors, a fire in a compartment that is on fire among the engine compartments; and [0012] following detecting a

fire in the compartment that is on fire, the method comprises an automatic assistance phase controlled by an avionics system, the automatic assistance phase comprising successively, automatically and with the avionics system, activating the fuel supply cut-off of the engine present in the compartment that is on fire so as to no longer supply fuel to this engine, then triggering the first extinguisher, then triggering the second extinguisher in predetermined respective conditions. [0013] Each engine compartment may be delimited by at least one firewall for preventing the detected fire from spreading to another zone of the aircraft.

[0014] For example, the engines are connected to a mechanical system. This mechanical system may possibly comprise a rotary wing, a propeller or indeed a yaw motion control rotor of the aircraft. For example, the aircraft is a rotorcraft, in particular a helicopter.

[0015] Therefore, if a fire is detected, the avionics system is configured to control the fuel supply cut-off when a first predetermined condition is met. This measure prevents the anti-fire agents emitted by the extinguishers from being sucked into the engine in the compartment that is on fire, optimizing the chances of extinguishing the fire. The avionics system is configured to then trigger the first extinguisher when a second predetermined condition is met, then the second extinguisher when a third predetermined condition is met.

[0016] The avionics system therefore helps to considerably reduce the workload of the pilot because, in the event of fire being detected, this avionics system takes various measures, automatically, to extinguish the detected fire. The pilot can therefore focus on other tasks, like searching for a landing area, for example. The avionics system also ensures that the fire extinguishing system of the aircraft is activated in the event of a fire being detected in a compartment that is on fire, which is not the case with a manual procedure that is subject to human error.

[0017] Furthermore, this method goes against the concerns of pilots, who want to be able to control a fire extinguishing system in order to prevent from extinguishers being triggered unnecessarily. However, as the aircraft is a multi-engine aircraft, even if one engine is unnecessarily switched off because of a fire being detected incorrectly, the aircraft still always comprises at least one engine in operation to ensure that the flight is completed safely.

[0018] Moreover, each fire detector may in particular comprise a thermocouple, thermistor or gas fire detector.

[0019] For example, a thermocouple fire detector may comprise two different metal blades that become deformed under the effect of an increase in temperature and move away from each other, opening an electric circuit when a detection threshold is reached. These two blades may comprise one blade that is deformed quickly and one blade that is deformed slowly.

[0020] A thermistor fire detector may comprise a temperature sensor based on the variation in an electrical resistance as a function of the temperature.

[0021] A gas fire detector may comprise a tube, for example made from stainless steel, filled with a gas-absorbing material. The increase in temperature that results, in particular, from fire, causes the gas in the sealed tube to be expelled, resulting in a rapid and detectable increase in pressure in the tube.

[0022] Such a fire detector is highly reliable. Arranging such fire detectors in a multi-engine aircraft that has an avionics system applying the method of the disclosure makes it possible to obtain a robust and reliable anti-fire method.

[0023] The method may also comprise one or more of the following features.

[0024] According to a first alternative, the aircraft being able to comprise at least one flight control member influencing the movement of the aircraft in the air, following detecting a fire in the compartment that is on fire, the automatic assistance phase comprises regulating a position of said at least one flight control member with the avionics system in order to place said aircraft in a predetermined flight configuration, activating the fuel supply cut-off being triggered at the same time as or after said regulation.

[0025] The avionics system is configured to then automatically place the aircraft in a particular flight configuration, compatible with the use of a single engine, for example. Therefore, the avionics system plays an active role in ensuring flight safety.

[0026] In this case, the first condition authorizing the activation of the fuel supply cut-off may include the detection of a fire and the meeting of the predetermined flight condition.

[0027] Said predetermined flight configuration may possibly be defined by at least one of the following parameters: a forward speed of the aircraft, an operating parameter of an engine different from the engine present in the compartment that is on fire, an altitude or a height of the aircraft. The operating parameter may be a driving power or an engine torque produced by a member of an engine that is not concerned by the detected fire, or by a member set in motion by this engine.

[0028] Regulating a position of said at least one flight control member with the avionics system may possibly comprise slaving at least a value of one of the following parameters to a predetermined respective setpoint value: a forward speed of the aircraft, a value of an operating parameter of an engine different from the engine present in the compartment that is on fire, an altitude or a height of the aircraft.

[0029] According to a second alternative, activating the fuel supply cut-off is triggered following detecting of a fire.

[0030] In this case, the supply of fuel to the engine present in the compartment that is on fire is cut off without any condition other than a fire being detected by a fire detector, or following a predetermined waiting period that gives a pilot the option of cancelling the automatic assistance phase by operating a human-machine shutdown interface. In this case, the first condition authorizing the activation of the fuel supply cut-off may include only the detection of a fire, and also, possibly, the expiry the predetermined waiting period. This predetermined waiting period may, for example, give a pilot the option of cancelling the automatic assistance phase.

[0031] According to one possibility compatible with the preceding possibilities, the automatic assistance phase may comprise measuring, with the avionics system, a speed of a movable member of the engine arranged in the compartment that is on fire, and triggering the first extinguisher, controlled by the avionics system, following activating the fuel supply cut-off when said speed of the movable member becomes less than or equal to a predetermined speed threshold.

[0032] For example, the engines are turboshaft engines. Therefore, the avionics system may comprise sensing devices measuring the speeds of rotation of the rotating assemblies of the gas generators of the turboshaft engines.

[0033] This feature helps maximize the chances of extinguishing the fire by reducing the risk of the anti-fire agent emitted by the extinguishers being sucked into the engine in question.

[0034] In this case, the second condition authorizing the triggering of the first extinguisher involves two factors: detecting when the fuel supply cut-off is activated, and detecting when the speed of the movable member of the engine present in the compartment that is on fire becomes less than or equal to a predetermined speed threshold.

[0035] Alternatively, the second condition authorizing the triggering of the first extinguisher may comprise only detecting when the fuel supply cut-off is activated, and possibly the expiry of a predetermined period starting from this activation.

[0036] According to one possibility compatible with the preceding possibilities, triggering the second extinguisher may be controlled by the avionics system if a fire detector still detects a fire after a predetermined monitoring period following triggering the first extinguisher.

[0037] For example, such a period is in the region of 5 seconds. The avionics system is configured to consider that, if a fire is still detected in the compartment that is on fire at the end of this monitoring period, the second extinguisher should be triggered to extinguish this fire.

[0038] The third condition authorizing the triggering of the second extinguisher may then be detecting, at the end of the monitoring period following the triggering of the first extinguisher, of a fire in the compartment that is on fire.

[0039] According to one possibility compatible with the preceding possibilities, the automatic assistance phase may comprise the following steps: [0040] when, at the end of a first predetermined period after triggering the first extinguisher, no fire detector detects a fire, the method comprises issuing a first alarm, the first alarm signaling at least the end of the automatic assistance phase or an order to land as soon as possible; [0041] when, at the end of the first predetermined period after triggering the first extinguisher, a fire detector detects a fire in the compartment that is on fire, the method comprises triggering the second extinguisher at the command of the avionics system; [0042] when, at the end of a second predetermined period after triggering the second extinguisher, no fire detector detects a fire, the method comprises issuing the first alarm; and [0043] when, at the end of the second predetermined period after triggering the second extinguisher, a fire detector detects a fire in the compartment that is on fire, the method comprises issuing a second alarm signaling an order to land immediately.

[0044] Depending on the seriousness of the current situation, an order to land immediately or as soon as possible may be issued by the avionics system, this order being intended for the pilot.

[0045] The order to land immediately is therefore issued if the two extinguishers have not succeeded in extinguishing the fire.

[0046] According to one possibility compatible with the preceding possibilities, following detecting a fire, the automatic assistance phase comprises generating, controlled by the avionics system and with an alerter, of a fire alert indicating the engine compartment concerned.

[0047] A pilot is thus informed that a fire has been detected and can act accordingly.

[0048] According to one possibility compatible with the preceding possibilities, following detecting a fire, the automatic assistance phase comprises generating, controlled by the avionics system and with an alerter, of an information alert following the launching of the automatic assistance phase.

[0049] A pilot is thus informed that the automatic assistance phase is under way. This step allows the pilot to focus on other tasks.

[0050] According to one possibility compatible with the preceding possibilities, following activating the fuel supply cut-off, the automatic assistance phase comprises generating, controlled by the avionics system and with an alerter, a status alert following said activation of the fuel supply cut-off.

[0051] A pilot is thus informed of the progress of the automatic assistance phase, and in particular of the cutting off of the supply of fuel to the engine concerned. This cut-off may be detected by the avionics system by means of a signal issued by a position sensing device measuring the position of a shutter of the fuel shut-off valve, or, for example, by a flowmeter measuring the flow rate of fuel transmitted to the engine.

[0052] The term “signal” refers hereinafter to an analog, digital, electrical or optical signal, for example.

[0053] According to one possibility compatible with the preceding possibilities, following triggering the first extinguisher, the automatic assistance phase comprises generating, controlled by the avionics system and with an alerter, a first extinguishing alert following this triggering of the first extinguisher.

[0054] When an extinguisher ejects the anti-fire agent into an engine compartment, the pressure in the vessel containing this agent drops. For example, each extinguisher may comprise a pressure sensor that transmits a signal when the pressure in the vessel of an extinguisher reaches a low threshold. The avionics system deduces from this that the extinguisher has functioned correctly and transmits a signal to the alerter.

[0055] A pilot is thus informed that the automatic assistance phase is under way and has caused the first extinguisher to be triggered.

[0056] According to one possibility compatible with the preceding possibilities, following triggering the second extinguisher, the automatic assistance phase may comprise generating,

controlled by the avionics system and with an alerter, a second extinguishing alert carrying this triggering of the second extinguisher.

[0057] A pilot is thus informed that the automatic assistance phase is under way and has caused the second extinguisher to be triggered.

[0058] According to one possibility compatible with the preceding possibilities, the method may comprise stopping the automatic assistance phase following the operation of a human-machine shutdown interface.

[0059] The pilot may at any time activate the human-machine shutdown interface if he or she deems it necessary in light of the various information received.

[0060] According to one possibility compatible with the preceding possibilities, the automatic assistance phase is activated only during flight by the avionics system. For example, the avionics system applies the automatic assistance phase when the height or altitude of the aircraft is greater than a threshold. To this end, the aircraft may comprise a conventional height or altitude-sensing device.

[0061] The disclosure also relates to a computer program comprising instructions that, when said program is run by the avionics system, cause said avionics system to implement the abovementioned method.

[0062] The disclosure also relates to an aircraft comprising at least two engines, each engine being arranged in its own engine compartment, each engine being connected to a fuel supply circuit provided with its own fuel supply cut-off for cutting off a fuel supply, said aircraft comprising at least one fire detector in each engine compartment, said aircraft comprising at least one first extinguisher and one second extinguisher.

[0063] This aircraft comprises an avionics system in communication with each fire detector, each fuel supply cut-off, and said at least one first extinguisher and one second extinguisher, in order to implement the abovementioned method.

[0064] The aircraft may possibly comprise at least one of the following features: each fuel supply cut-off is a fuel shut-off valve, each of said at least one first extinguisher and one second extinguisher comprises an anti-fire agent and one pipe for each engine compartment for conveying said anti-fire agent into each engine compartment.

[0065] Each fire detector may possibly comprise a thermocouple, thermistor or gas fire detector.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0066] The disclosure and its advantages appear in greater detail in the context of the following description of embodiments given by way of illustration and with reference to the accompanying figures, wherein:

[0067] FIG. 1 is a schematic view of an aircraft according to the disclosure; and

[0068] FIG. 2 is a logic diagram showing the method of the disclosure.

DETAILED DESCRIPTION

[0069] Elements that are present in more than one of the figures are given the same references in each of them.

[0070] FIG. 1 shows an aircraft 1 capable of implementing the method of the disclosure described hereinafter.

[0071] This aircraft 1 comprises at least two engines 16, 17. Reference number 15 is used to denote any engine, if required. The engines 15 may be connected to a mechanical system 2. For example, this mechanical system 2 comprises a power transmission system 5 that sets a rotating system 7 in motion. According to the example shown, the power transmission system 5 comprises a gearbox 6 connected by conventional members, which are not shown here in order not to clutter the Figure, to

each engine **15** and to the rotating system **7**. This rotating system **7** may comprise at least one blade **8**, possibly carried by a hub **9**. Such a rotating system **7** may form a propeller, a rotary wing or a yaw motion control rotor, to give various examples.

[0072] Irrespective of this aspect, each engine **15** may be an engine supplied with fuel. Each engine **16,17** is then connected to a fuel supply circuit **31, 36**. Each fuel supply circuit **31, 36** comprises at least one tank **33, 330**, possibly shared with the other fuel circuit, at least one duct **32, 37** linking the tank to the associated engine, and at least one fuel supply cut-off **34, 38** specific to the associated engine **15**. Therefore, the first engine **16** is supplied with fuel by a first fuel supply circuit **31**, this first fuel supply circuit **31** comprising a first duct **32** equipped with at least one first fuel supply cut-off **34** and extending from a tank **33** to the first engine **16**. Similarly, the second engine **17** is supplied with fuel by a second fuel supply circuit **36**, the second fuel supply circuit **36** comprising a second duct **37** equipped with at least one second fuel supply cut-off **38** and extending from a tank **330** to the second engine **17**.

[0073] Each fuel supply cut-off **34, 38** may comprise a fuel shut-off valve, and/or a pump. Activating the fuel supply cut-off therefore involves closing the fuel shut-off valve or stopping the pump, depending on the variant.

[0074] Moreover, each engine **16,17** comprises a movable member **151, 152**. In the context of a turboshaft engine, this movable member **151, 152** may be the rotating assembly of the gas generator, and may comprise at least one compression stage constrained to rotate with at least one turbine.

[0075] Moreover, each engine **16, 17** is arranged in its own engine compartment **21, 22**. Reference number **20** denotes any engine compartment, if required. Each engine compartment **21,22** may be delimited by at least one firewall for containing a possible fire in this engine compartment **21,22**.

[0076] Furthermore, at least one fire detector **26-29** is housed in each engine compartment **21, 22**. Reference number **25** may denote any fire detector, if required. For example, a given engine compartment **21, 22** may comprise fire detectors **26-27, 28-29** set at different triggering temperatures depending on their positioning. Each fire detector **26-29** may comprise a thermocouple, thermistor or gas fire detector.

[0077] Moreover, the aircraft **1** comprises at least one first extinguisher **40** and one second extinguisher **45**. Each extinguisher **40, 45** comprises a vessel housing an anti-fire agent, such as a halon. Furthermore, each extinguisher **40, 45** may comprise one pipe for each engine compartment **21,22**, i.e., a first pipe **41, 46** leading into the first engine compartment **21** and a second pipe **43, 48** leading into the second engine compartment **22** according to the example shown. Each extinguisher **40,45** comprises one trigger **42, 44, 47, 49** per pipe **41, 43, 46, 48** of this extinguisher **40,45**, such as pyrotechnic cartridges, for example, for conveying the anti-fire agent to one or another of the pipes **41, 43, 46,48**.

[0078] Alternatively, each engine compartment may have its own extinguishers, for example.

[0079] Furthermore, each extinguisher **40, 45** may comprise a pressure sensor **400, 500** that transmits a particular signal when the pressure inside its vessel reaches a low threshold that signifies the injection of the anti-fire agent into an engine compartment.

[0080] Moreover, the aircraft **1** comprises an avionics system **60** in communication with each fire detector **26-29**, each fuel supply cut-off **34, 38**, and each extinguisher **40, 45** in order to implement the method of the disclosure.

[0081] For example, this avionics system **60** may comprise one speed sensing device **18, 19** per engine **16,17** measuring the speed of the movable member **151, 152** of this engine **16, 17**, i.e., a first speed sensing device **18** for the first engine **16** and a second speed sensing device **19** for the second engine **17**.

[0082] The avionics system **60** may comprise a controller in communication with the speed sensing devices **18, 19**. The controller may run a computer program comprising instructions that, when the program is run by the avionics system **60**, causes the avionics system **60** to implement the method

of the disclosure.

[0083] This controller may comprise one engine computer per engine **15**, i.e., a first engine computer **160** controlling the first engine **16** and a second engine computer **170** controlling the second engine **17**, according to the example shown.

[0084] This controller may comprise an automatic flight control computer **65**, possibly in communication with the engine computers **160**, **170**.

[0085] By way of example, each described computer may comprise at least one processor and at least one memory, at least one integrated circuit, at least one programmable system, or at least one logic circuit, these examples not limiting the scope to be given to the term “computer”. The term “processor” may refer equally to a central processing unit or CPU, a graphics processing unit or GPU, a digital signal processor or DSP, a microcontroller, etc.

[0086] According to another aspect, the avionics system **60** may be configured to automatically control the aircraft **1** in certain conditions, by controlling a position of one or more flight control members. The controller may, for example, run a program for this purpose.

[0087] Such a flight control member may be in the form of a blade **8** according to the example shown, the pitch angle of the blade **8** being capable of being controlled using one or more servocontrols **88**. Another flight control member may be in the form of a fuel metering valve supplying fuel to an engine **15**, a blade of another rotor, a vertical stabilizer flap or tailplane flap, etc. Irrespective of the nature of the flight control member, the position of the flight control member can be controlled with an actuator of the avionics system **60**.

[0088] For example, each blade **8** may be hinged to a pitch rod **90** that is hinged to a set of swashplates **89**, the position of the set of swashplates **89** being capable of being controlled with servocontrols **88**. Each servocontrol **88** may be controlled by a control channel **86** comprising at least one actuator **87**. The controller, and possibly the automatic flight control computer **65**, may control the actuators **87**.

[0089] Moreover, the avionics system **60** may comprise a human-machine control interface **95** for requesting the application of the method of the disclosure, and/or a human-machine shutdown interface **96**. These interfaces transmits signals to the controller, or to the automatic flight control computer **65**, in particular. Each human-machine interface may comprise at least one button, a touch screen, a mouse, a keyboard, a voice system, etc.

[0090] Moreover, the aircraft **1** may comprise an alerter **70** capable of generating various alerts at the command of the controller and, for example, of the automatic flight control computer **65**. Each alert may be in the form of a visual alarm, for example by displaying one or more characters on a screen, and/or an audible and/or haptic alarm. If there is a visual alarm, the alerter **70** may comprise one or more screens. The controller, or the automatic flight control computer **65**, may transmit one or more signals to the alerter in order to request the issuing of the required alerts.

[0091] Moreover, the aircraft **1** may comprise a height- or altitude-sensing device **97** measuring a height or an altitude, for example a radio altimeter, or a conventional forward speed sensing device **970** measuring a speed of the aircraft and, for example, an air speed.

[0092] The aircraft **1** may possibly comprise a position sensing device **340**, **380** in each fuel supply cut-off **34**, **38** for assessing whether a member of this fuel supply cut-off is in a closed position, or a flow rate sensing device **341**, **381** in each fuel supply circuit **31**, **36** for determining whether the supply of fuel to an engine **15** is open or closed.

[0093] These sensing devices **97**, **970**, **341**, **381** transmit signals to the controller, or to the automatic flight control computer **65**, in particular.

[0094] FIG. **2** shows a method for controlling such an aircraft **1** in the event of a fire in an engine compartment.

[0095] This method may comprise a step of activating the method by operating the human-machine control interface **95**.

[0096] In the event of a fire, the method comprises detecting, during a step STP**1**, with at least one

of the fire detectors **26-29**, a fire in an engine compartment referred to as the “compartment that is on fire”. To illustrate the disclosure, the compartment that is on fire will be considered hereinafter to be the first engine compartment **21**.

[0097] A fire detector, for example the detector **27**, then transmits a signal to the controller of the avionics system **60** and, for example, to the automatic flight control computer **65**. The controller decodes the signal and deduces therefrom that there is a fire in the compartment that is on fire **21**.

[0098] Following this detection STP**1** of a fire in the compartment that is on fire **21**, an automatic assistance phase PHASASSIST is controlled by the avionics system **60**. The avionics system **60** possibly activates the automatic assistance phase PHASASSIST only when the aircraft is at a height or an altitude, measured with the sensing device **97**, higher than a threshold.

[0099] During this automatic assistance phase PHASASSIST, the avionics system **60** controls, successively, automatically and in predetermined conditions, the activation STP**5** of the fuel supply cut-off **34** of the engine **16** present in the compartment that is on fire **21**, then the triggering STP**7** of the first extinguisher **40**, then the triggering STP**9** of the second extinguisher **45**, and possibly the issuing of various alerts.

[0100] The method may possibly comprise stopping the automatic assistance phase PHASASSIST at any time after operating the human-machine shutdown interface **96**. This human-machine shutdown interface **96** then transmits a signal to the controller that stops the automatic assistance phase. The fuel supply cut-offs **34**, **38** and the extinguishers **40**, **45** may then be controlled manually with conventional dedicated controls.

[0101] According to one possibility, following the detection STP**1** of a fire, the method comprises generating STP**2** a fire alert signaling the engine compartment concerned. After receiving a fire detection signal from a fire detector, the avionics system **60** controls the alerter **70** to generate a fire alert signaling the detection of a fire and the engine compartment concerned. For example, the controller, or the automatic flight control computer **65**, transmits a signal to the alerter **70** to request the generation of the fire alert. This fire alert specifies the engine compartment wherein the fire has been detected. For example, the message “ENG1 FIRE” is displayed to signal a fire in the first engine compartment.

[0102] Alternatively, afterwards or at the same time, the automatic assistance phase PHASASSIST may comprise generating STP**3** an information alert carrying the launching of the automatic assistance phase PHASASSIST. For example, the controller, or the automatic flight control computer **65**, transmits a signal to the alerter **70** to request the generation of the information alert, after receiving a fire detection signal from a fire detector. For example, the message “ENG1 FIRE AUTO PROT” is displayed.

[0103] Alternatively, after or at the same time as the preceding alert steps, following the detection STP**1** of a fire, the automatic assistance phase PHASASSIST may comprise regulating STP**4** a position of at least one flight control member in order to position the aircraft **1** in a predetermined flight configuration, the activation STP**5** of the fuel supply cut-off **34**, **38** being triggered at the same time as or after said regulation STP**4**.

[0104] For example, after receiving a fire detection signal from a fire detector, the controller, or the automatic flight control computer **65**, transmits a signal to at least one actuator acting on a position of a flight control member in order to slave at least a value of one of the following parameters to a predetermined setpoint value: a forward speed of the aircraft **1** measured with the conventional forward speed sensing device **970**, a value of an operating parameter of an engine different from the engine present in the compartment that is on fire measured with the speed sensing device **19**, an altitude or a height of the aircraft **1** measured with the conventional forward speed sensing device **970**.

[0105] Irrespective of these possibilities, the activation STP**5** of the fuel supply cut-off **34** may be triggered following the detection STP**1** of a fire. The controller, or the automatic flight control computer **65**, transmits a signal to the fuel supply cut-off **34** that cuts off the supply of fuel to the

first engine **16** according to the example shown, for example after receiving a fire detection signal from a fire detector.

[0106] Possibly, after activating STP5 the fuel supply cut-off **34**, the automatic assistance phase PHASASSIST comprises generating STP6 a status alert. The status alert carries a status signaling that the fuel supply has or has not been cut off. For example, a sensing device for sensing the position of the fuel shut-off valve or a flow rate sensing device transmits a signal to the controller to inform it that the supply of fuel to the engine present in the compartment that is on fire has been cut off. The controller, or the automatic flight control computer **65**, then transmits a signal to the alerter **70** to request the generation of the status alert. For example, the message “ENG1 OFF” is displayed, signaling that the fuel shut-off valve is closed.

[0107] For example, if a pilot notices that the status alert has not been issued, the pilot can cancel the automatic assistance phase during a stop phase STPOFF, and/or can close off the fuel supply by another means during a step STPMAN and thus resume the automatic assistance phase.

[0108] Irrespective of these options, after closing off the supply of fuel to the engine **16** present in the compartment that is on fire **21**, the avionics system **60** and, for example, the controller, or the automatic flight control computer **65**, transmits a signal to the first extinguisher **40** to trigger STP7 the first extinguisher **40**. The trigger **42** is then actuated to inject the anti-fire agent of the first extinguisher **40** into the compartment that is on fire.

[0109] The automatic assistance phase PHASASSIST possibly comprises measuring STPCOND, with the avionics system **60**, and in particular with the speed sensing device **18** according to the example given, a speed of a movable member **151** of the engine **16** of the compartment that is on fire **21**. The avionics system **60** then controls the triggering STP7 of the first extinguisher **40** only when the speed of the movable member **151** becomes less than or equal to a predetermined speed threshold and a fire is detected.

[0110] According to one possibility, the automatic assistance phase PHASASSIST comprises generating STP8, with the alerter **70**, a first extinguishing alert carrying this triggering of the first extinguisher **40**. For example, the pressure sensing device **400** of the first extinguisher **40** transmits a signal to the controller and, for example, to the automatic flight control computer **65**, this controller consequently transmitting a signal to the alerter **70** to request the issuing of the first extinguishing alert. For example, the message “ENG1 FIRE SHOT 1” is displayed.

[0111] Moreover, the assistance phase PHASASSIST possibly comprises, after the triggering of the first extinguisher **40**, generating, with the alerter **70** and at the command of the avionics system **60**, a second alarm if at least one fire detector **26-29** detects a fire or a first alarm if no fire detector **26-29** detects a fire. The second alarm signals an order to land immediately, the first alarm signaling at least the end of the automatic assistance phase or an order to land as soon as possible. For example, the first alarm comprises the message “LAND AS SOON AS POSSIBLE”, and the second alarm comprises the message “LAND IMMEDIATELY”.

[0112] For example, the automatic assistance phase PHASASSIST comprises the following steps.

[0113] Thus, if, at the end of a first predetermined period after the triggering STP7 of the first extinguisher **40**, no fire detector **26-29**, in particular that of the compartment that is on fire **21**, transmits a signal to the controller to signal a fire, the fire is considered to be extinguished. The method comprises issuing STP10 the first alarm. For example, the controller, or the automatic flight control computer **65**, transmits a signal to the alerter **70** to this effect in order to request the issuing of the first alarm. The pilot is then informed that the automatic assistance phase PHASASSIST is complete and, therefore, that the fire has been extinguished and/or that it is advisable to land as soon as possible.

[0114] Conversely, if, at the end of a predetermined monitoring period after the triggering of the first extinguisher **40**, a fire detector **26-27** detects a fire in the compartment that is on fire **21**, the method comprises triggering STP9 the second extinguisher **45**. The monitoring period may be equal to or different from the first predetermined period. For example, the controller, or the

automatic flight control computer **65**, transmits a signal to the second extinguisher **45** to this effect. The trigger **47** is then actuated to inject the anti-fire agent of the second extinguisher **45** into the compartment that is on fire.

[0115] The automatic assistance phase PHASASSIST possibly comprises generating STP**11**, with the alerter **70**, a second extinguishing alert carrying this triggering STP**9** of the second extinguisher **45**. For example, the controller, or the automatic flight control computer **65**, transmits a signal to the alerter **70** to this effect. For example, the message “ENG1 FIRE SHOT 2” is displayed.

[0116] Thus, if at the end of a second predetermined period after the triggering STP**9** of the second extinguisher **45**, no fire detector **26-29**, in particular that of the compartment that is on fire **21**, transmits a signal to the controller to signal a fire, the fire is extinguished. The method comprises issuing STP**12** the first alarm.

[0117] The triggering STP**9** of the second extinguisher **45** is controlled by the avionics system **60** if a fire detector **26-27** still detects a fire at the end of the first predetermined activation period after the triggering of the first extinguisher **40**.

[0118] Conversely, if, at the end of the second predetermined period after the triggering STP**9** of the second extinguisher **45**, a fire detector **26-29** detects a fire, the method comprises issuing STP**13** the second alarm. For example, the controller, or the automatic flight control computer **65**, transmits a signal to the alerter **70** to this effect. The pilot is then informed that the automatic assistance phase PHASASSIST has not allowed the fire to be extinguished and that it is advisable to land immediately.

[0119] Naturally, the present disclosure is subject to numerous variations as regards its implementation. Although several embodiments are described above, it should readily be understood that it is not conceivable to identify exhaustively all the possible embodiments. It is naturally possible to replace any of the means described with equivalent means without going beyond the ambit of the present disclosure.

Claims

1. A method for controlling an aircraft comprising at least two engines, each engine being arranged in its own engine compartment, each engine being connected to a fuel supply circuit provided with its own fuel supply cut-off for cutting off a fuel supply, the aircraft comprising at least one fire detector in each engine compartment, the aircraft comprising at least one first extinguisher and one second extinguisher, wherein the control method comprises: detecting, with at least one of the fire detectors, a fire in a compartment that is on fire among the engine compartments; and following detecting a fire in the compartment that is on fire, the method comprises an automatic assistance phase controlled by an avionics system, the automatic assistance phase comprising successively, automatically and with the avionics system, activating the fuel supply cut-off of the engine present in the compartment that is on fire so as to no longer supply fuel to this engine, then triggering the first extinguisher, then triggering the second extinguisher in predetermined respective conditions.
2. The method according to claim 1, wherein, the aircraft comprising at least one flight control member influencing a movement of the aircraft in the air, following detecting a fire in the compartment that is on fire, the automatic assistance phase comprises regulating a position of the at least one flight control member with the avionics system in order to place the aircraft in a predetermined flight configuration, activating the fuel supply cut-off being triggered at the same time as or after the regulation.
3. The method according to claim 2, wherein the predetermined flight configuration is defined by at least one of the following parameters: a forward speed of the aircraft, an operating parameter of an engine different from the engine present in the compartment that is on fire, an altitude or a height of the aircraft.
4. The method according to claim 2, wherein regulating a position of the at least one flight control

member with the avionics system comprises slaving at least one value of one of the following parameters to a predetermined respective setpoint value: a forward speed of the aircraft, a value of an operating parameter of an engine different from the engine present in the compartment that is on fire, an altitude or a height of the aircraft.

5. The method according to claim 1, wherein activating the fuel supply cut-off is triggered following the detection of a fire.

6. The method according to claim 1, wherein the automatic assistance phase comprises measuring, with the avionics system, a speed of a movable member of the engine arranged in the compartment that is on fire, and triggering the first extinguisher controlled by the avionics system following activating the fuel supply cut-off when the speed of the movable member becomes less than or equal to a predetermined speed threshold.

7. The method according to claim 1, wherein triggering the second extinguisher is controlled by the avionics system if a fire detector still detects a fire at the end of a predetermined monitoring period after triggering the first extinguisher.

8. The method according to claim 1, wherein the automatic assistance phase comprises the following steps: when, at the end of a first predetermined period after triggering the first extinguisher, no fire detector detects a fire, the method comprises issuing a first alarm, the first alarm signaling at least the end of the automatic assistance phase or an order to land as soon as possible; when, at the end of the first predetermined period after triggering the first extinguisher, a fire detector detects a fire in the compartment that is on fire, the method comprises the triggering the second extinguisher at the command of the avionics system; when, at the end of a second predetermined period after triggering the second extinguisher, no fire detector detects a fire, the method comprises issuing the first alarm; and when, at the end of the second predetermined period after triggering the second extinguisher, a fire detector detects a fire in the compartment that is on fire, the method comprises issuing a second alarm signaling an order to land immediately.

9. The method according to claim 1, wherein, following detecting a fire, the automatic assistance phase comprises generating, controlled by the avionics system and with an alerter, a fire alert indicating the engine compartment concerned.

10. The method according to claim 1, wherein, following detecting of a fire, the automatic assistance phase comprises generating, controlled by the avionics system and with an alerter, an information alert following the launching of the automatic assistance phase.

11. The method according to claim 1, wherein, following activating the fuel supply cut-off, the automatic assistance phase comprises generating, controlled by the avionics system and with an alerter, of a status alert following the activation of the fuel supply cut-off.

12. The method according to claim 1, wherein, following triggering the first extinguisher, the automatic assistance phase comprises generating, controlled by the avionics system and with an alerter, a first extinguishing alert following this triggering of the first extinguisher.

13. The method according to claim 1, wherein, following triggering the second extinguisher, the automatic assistance phase comprises generating, controlled by the avionics system and with an alerter, a second extinguishing alert.

14. The method according to claim 1, wherein the method comprises stopping the automatic assistance phase after operating a human-machine shutdown interface.

15. An aircraft comprising at least two engines, each engine being arranged in its own engine compartment, each engine being connected to a fuel supply circuit provided with its own fuel supply cut-off for cutting off a fuel supply, the aircraft comprising at least one fire detector in each engine compartment, the aircraft comprising at least one first extinguisher and one second extinguisher, wherein the aircraft comprises an avionics system in communication with each fire detector and with each fuel supply cut-off and with the at least one first extinguisher and one second extinguisher in order to implement the method according to claim 1.

16. The aircraft according to claim 15, wherein the aircraft comprises at least one of the following

features: each fuel supply cut-off is a fuel shut-off valve, each of the at least one first extinguisher and one second extinguisher comprises an anti-fire agent and one pipe for each engine compartment for conveying the anti-fire agent into each engine compartment.

17. The aircraft according to claim 15, wherein each fire detector comprises a thermocouple, thermistor or gas fire detector.
